

Wearable Brain-Computer Interface (BCI)

TinkerHub Community Meetup — Event Report

EVENT NAME	Talk Session - Wearable Brain Computer Interface (BCI)
HOSTED BY	Sabarinath R
DATE	Saturday 15 August 2026
TIME	8:00pm - 9:00pm
VENUE	Online Mode (Google Meet)

EVENT OVERVIEW

The session on Wearable Brain-Computer Interface (BCI) introduced participants to the concept of connecting brain signals with technology. The session explored non-invasive BCI and how signals generated by the brain can be captured, processed, and converted into actions.

The session demonstrated the idea of using a wearable headband to enable hands-free interaction with technology. Instead of relying on conventional input devices such as a mouse or keyboard, brain signals can be used as an input, which can then be processed through software to perform different actions.

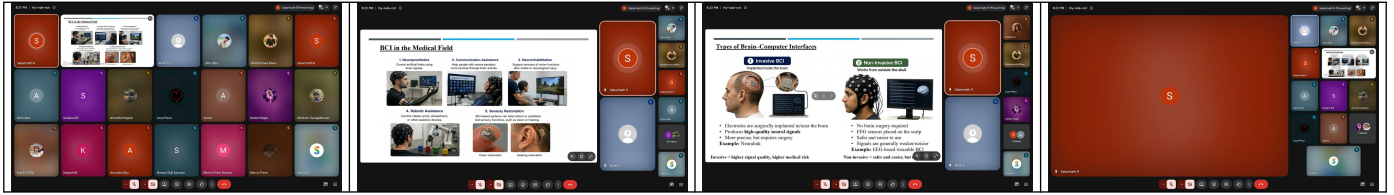
The session also explored multiple activities based on this concept, including hands-free navigation and gaming, giving participants an insight into how brain signals can potentially be transformed into real-time interactions with technology.

"Your thoughts. Your signals. Your future"

OBJECTIVES

- To introduce participants to the concept of Brain-Computer Interfaces (BCI).
- To explore non-invasive and wearable BCI technology.
- To understand how brain signals can be captured and processed.
- To explore how brain signals can be converted into actions.
- To demonstrate the possibilities of hands-free interaction with technology.
- To provide an insight into the potential applications of wearable BCI systems.

EVENT HIGHLIGHTS



OUTCOME

- **Understanding BCI Technology:** Participants gained an introductory understanding of wearable Brain-Computer Interfaces and their role in connecting brain signals with technology.
- **Brain Signals as Input:** The session helped participants understand how signals generated by the brain can be captured and used as an input for technological systems.
- **Signal-to-Action Interaction:** Participants learned how brain signals can be processed and translated into actions, demonstrating a new approach to interacting with technology.
- **Hands-Free Interaction:** The demonstrations showcased the possibilities of hands-free navigation and interaction using a wearable BCI headband.
- **Exploring Future Applications:** Through multiple activities and demonstrations, participants gained an insight into the potential applications of BCI technology in areas such as gaming and interactive systems.

CONCLUSION

The Wearable Brain-Computer Interface session provided an engaging introduction to the possibilities of connecting brain signals with technology. Through demonstrations and activities, participants gained an understanding of how signals from the brain can be captured, processed, and transformed into actions. The session offered a glimpse into the future of hands-free interaction and the potential of BCI technology to change the way we interact with digital systems.